

Part B → System design ScenariosQuestion 1 → Hotel Booking System* Algorithm

① Start

② Total revenue = 0, guest count = 1

③ Input N

④ While guestcount $\leq$ N

Input Season

Input room type

Input nights

If season == 1

If room type == 1

rate = 5000

Else If room type == 3

rate = 12000

 Else If ~~season == 2~~ ^{Else 26} If room type == 2

rate = 8000

Else If season == 2

If room type == 1

rate = 3000

Else If room type == 2

rate = 5000

Else If room type == 3

rate = 8000

base price = rate * nights

If nights > 7

discount = base price * 0.15

Else

discount = 0



guesttotal = base price - discount

Print guesttotal

Total revenue = Total revenue + guest total

guest count = guest count + 1

END While

Print Total revenue

~~END~~

(4) END

* * Pseudocode

Begin

Set total revenue = 0

Set guest count

Print "Enter total number of guests"

Input N

While guestcount $\leq$ N

Print "Guest are", guestcount

Print "Enter Season (1 for Peak, 2 for Off-Peak)"

Input Season

Print "Enter room type (1 for standard, 2 for Deluxe, 3 suite)"

Input room type

Print "Enter number of nights stayed"

Input nights

IF season == 1 &

 If roomtype == 1

 Set rate = 5000

 Else If room-type == 2

 Set rate = 8000

 Else If roomtype == 3

 Set rate = 12000

~~END~~ END IF



Else If $\text{room} == 2$

 If $\text{roomtype} == 1$

 set rate = 3000

 Else If $\text{roomtype} == 2$

 set rate = 5000

 Else If $\text{roomtype} == 3$

 set rate = 8000

 EndIf

EndIf

set base price = rate * nights

If nights > 7

 set discount = base price * 0.15

Else

 set discount = 0

EndIf

Set guest total = base price - discount

print "Total Bill for Guest", guest total

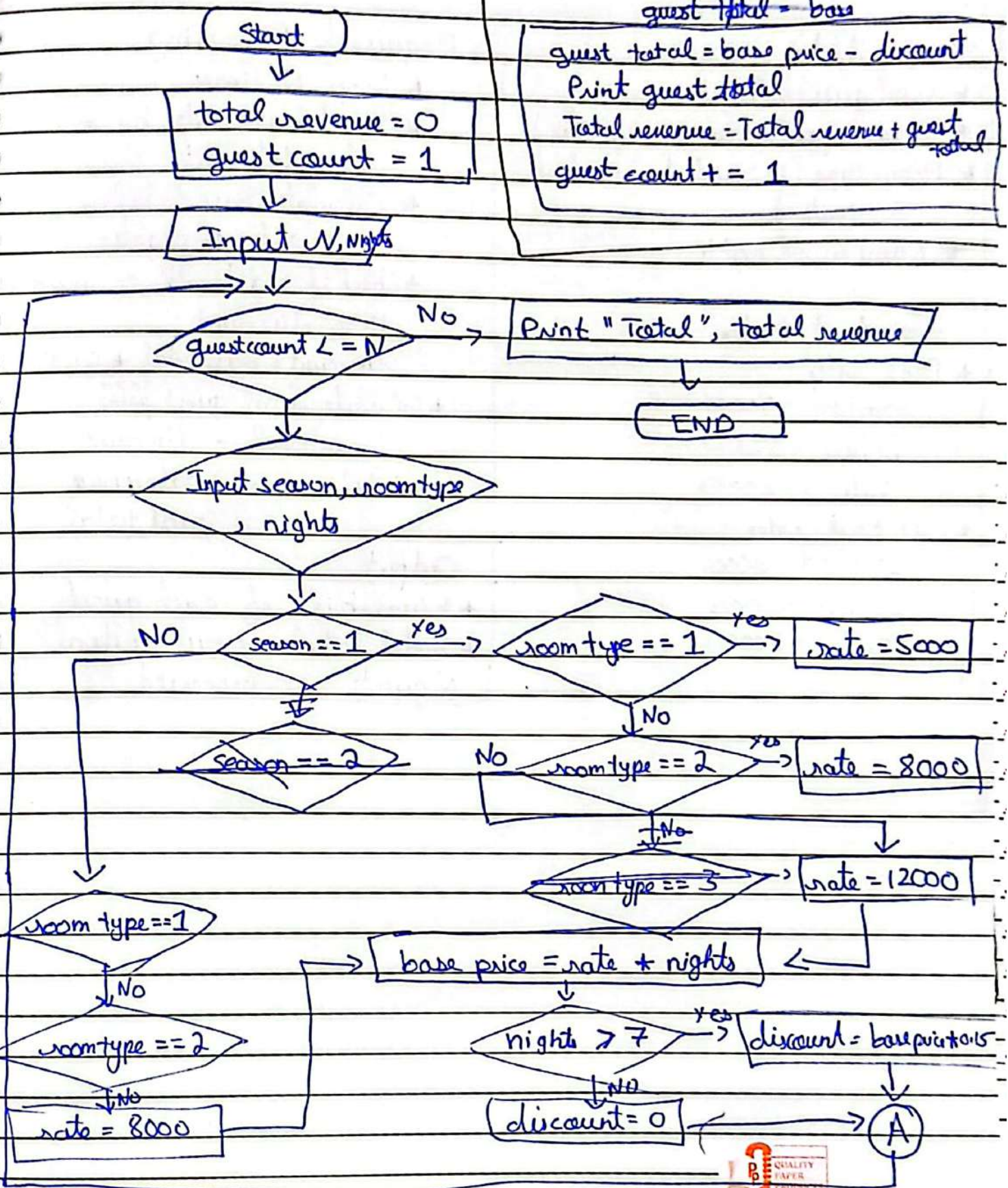
Set total revenue = total revenue + guest total

Set guest count = guest count + 1

ENDWhile

Print "Hotel Total Revenue: ", total revenue

* Flowchart



* Problem Analysis Chart (PAC)

Given Data	Required Processing
* Total guests (N)	* Loop N times
* Season type (1=Peak, 2=Off Peak)	* Select base rate by If else statement
* Room type (1=Standard, 2=Deluxe, 3=Suite)	* Calculate base total = rate * nights
* Number of nights	* Check if nights > 7 so apply 15% discount discount = base total * 0.15
Constant/Rates	* Calculate final guest ^{total} price = base total - discount
* Peak rates Standard = 5000 Deluxe = 8000 Suite = 12000	* total revenue = total revenue + guest total
* Off-Peak rates Standard = 3000 Deluxe = 5000 Suite = 8000	Output * Final bill of each guest * Hotel total revenue after all N guests are processed

2
0.5
0.5
2.0

Date: _____ 20____

* IPO Chart

INPUT	Processing	Output
* N (number of guests)	(1) Initializing	* guest total (for each guest)
* Season (1 for Peak 2 for Off Peak)	* guest count = 1 * total revenue = 0	* total revenue
* room type (1 for standard 2 for Deluxe, 3 for Suite)	(2) Loop	
* nights (number of nights stayed)	* While (guest count $\leq$ N)	
	(a) Read guest details (Season, room type, nights)	
	(b) Determine room rate based on season and room type	
	(c) Calculate base price = rate * nights	
	(d) If nights > 7 set discount = base price * 0.15 else discount = 0	
	(e) Calculate guest total = base price - discount	
	(f) Print guest total	
	(g) Update total revenue + = guest total	
	(3) Print Total revenue	

